

A formal counterexample: mixed ratios evaluate on the face, not at the corner

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Abstract

Unit 206 records, as theorems, the smallest instance of the paper-facing correction reported since unit 186: for *mixed* exponent ratios the leading coefficient of a dressed normal moment integral is the face functional of the amplitude, not its value at the origin. With $d = 2$, $h = (0, 2)$, $k = (1, 1)$, $\beta = 1$ (ratios $\frac{1}{2}, \frac{3}{2}$; $\lambda = \frac{1}{2}$, $J = \{0\}$) and $\eta(u) = 1 + u_1$,

$$\frac{\int_{(0,1)^2} (1 + u_1) u_1^2 e^{-Nu_0^2 u_1^2} du}{N^{-1/2}} \rightarrow \frac{5\sqrt{\pi}}{12}, \quad \eta(0) \cdot \frac{\Gamma(\frac{1}{2})}{2(2+1-1)} = \frac{\sqrt{\pi}}{4},$$

and $5\sqrt{\pi}/12 \neq \sqrt{\pi}/4$ (`MixedCounterexample.tendsto`, `corner_evaluation_fails`). Review v10 (units 204–205) reported no statement-level defects; Astra #22 ranks the remaining work B2 (finite chart assembly) $\rightarrow$ A2 (general- d stochastic transfer by equi-Lipschitz control) $\rightarrow$ D2 (signed orthants with phase, where the phase reflects with $\varepsilon_\sigma = \prod_i \sigma_i^{k_i}$) $\rightarrow$ hand-off; C2 stays NO-GO. Zero sorry/axiom; 9083 jobs.

1 The things formalised

```
namespace MixedCounterexample
def hEx : Fin 2 → ℝ := ![0, 2]; def kEx : Fin 2 → ℝ := ![1, 1]
noncomputable def Ex (u : Fin 2 → ℝ) : ℝ := 1 + u 1
theorem ratio0 : ratioExp hEx kEx 0 = 1/2; theorem ratio1 : ratioExp hEx kEx 1 = 3/2
theorem mixedConst_eq : monomialMixedConst hEx kEx (1/2) 1 = √ / 4
theorem mixedConst_shift_eq :
  monomialMixedConst (fun i => hEx i + ![0, 1] i) kEx (1/2) 1 = √ / 6
theorem amplitudeCoeff_eq : amplitudeCoeff hEx kEx (1/2) 1 Ex = 5 * √ / 12
theorem corner_eq : Ex 0 * monomialMixedConst hEx kEx (1/2) 1 = √ / 4
theorem corner_evaluation_fails :
  amplitudeCoeff hEx kEx (1/2) 1 Ex Ex 0 * monomialMixedConst hEx kEx (1/2) 1
theorem tendsto : Tendsto (fun N => ( x in unitBox 2, Ex x * (( i, x i ^ hEx i)
  * exp (-(1 * N * i, x i ^ (2 * kEx i)))))) / N ^ (-(1/2 : ℝ))) atTop ( (5 * √ / 12))
```

2 Mathematics

No integration is performed by hand: the amplitude $1 + u_1$ is the sum of the two monomials $u^{(0,0)}$ and $u^{(0,1)}$, and the face-moment theorem of unit 186 (`face_moment_eq`: $\int (\pi u)^\gamma \text{faceLeadConst } w = \text{monomialMixedConst}(h + \gamma)$ when γ vanishes on J) evaluates each as a bare mixed constant, $\sqrt{\pi}/4$ for $h = (0, 2)$ and $\sqrt{\pi}/6$ for $h = (0, 3)$, via the two-dimensional closed form of unit 198. The sum $5\sqrt{\pi}/12$ is the face value $\frac{\sqrt{\pi}}{2} \int_0^1 (1+v)v dv$; the corner value $\sqrt{\pi}/4$ forgets the u_1 -integration against the residual weight $u_1^{h_1-2k_1\lambda} = u_1$.

3 Lean notes

- `simp [ratioExp, hEx, kEx]` closes $(0 + 1)/(2 \cdot 1) = 1/2$ outright but leaves $2 + 1 = 3$ for $(2 + 1)/(2 \cdot 1) = 3/2$; a trailing `norm_num` is needed in one case and is an error (“no goals”) in the other. Check each.
- A hypothesis stated for the abbreviations `hEx kEx` is accepted by `defeq` as a term argument but not by `rw` inside a goal stated with `![0,2] ![1,1]`; use `show ratioExp hEx kEx 1 1/2` first.
- `Real.Gamma_one_half_eq` : $\Gamma(1/2) = \sqrt{\pi}$ and `Real.one_rpow` reduce the closed forms; `nlinarith` with $0 < \sqrt{\pi}$ separates $5\sqrt{\pi}/12$ from $\sqrt{\pi}/4$.

4 Review v10 and Astra #22

Review v10 (units 204–205): clean; mirror should say “ $c > 0$ on the closed cube” with continuity, keep the “leading quotient limit, not full expansion” caveat, and make clear that the cutoff face integral is the unit-box parametrisation (the b -factor is displayed once). Astra #22: next is B2, the conditional finite chart assembly (minimum exponent wins, then *maximum* log multiplicity; assemble numerator and denominator separately; 2–4 units), then A2 (general- d stochastic transfer via eventual equi-Lipschitz dependence of the normalised integral on (η, ξ) in sup norm, 4–8 units), then D2 if the paper integrates full signed fibres (the phase reflects: $\xi_\sigma(u) = \varepsilon_\sigma \xi(\sigma u)$, $\varepsilon_\sigma = \prod \sigma_i^{k_i}$; odd monomials give $J_p(a) + J_p(-a)$, not $2J_p(a)$).